

JAIN (DEEMED-TO-BE UNIVERSITY)

BENGALURU, KARNATAKA

P2: Teaching-Learning and Evaluation Plan

PROGRAMME: BCA (Bachelor of Computer Applications) - CTIS

SEMESTER - IV

Teaching - Learning & Evaluation Plan

Course Information:

Course Code: **23BCA4C03**

Course Title: **PYTHON PROGRAMMING**

Credits Units: 03

Contact Hours : 45

L-T-P-E : 3-0-0-3

IA: ESE Weightage- 50 :50

Pass Marks (IA & ESE) – 40 (ESE – Min.18)

Aggregate Pass Marks: 40%

ESE Question Paper Marks: 50

Special Examination Fees: NA

Course Facilitator (s): Dr.A.Kannagi , Associate Professor , School of CS & IT

Dr. Saumya Patel, Assistant Professor, School of CS & IT

Outcomes (POs) and Programme Specific Outcomes (PSOs)

Programme Outcomes (POs)	
At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.

PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.

Program Specific Outcomes (PSO's)

PSO 1	Explore the impact of cloud and security in a global scenario and be a successful computer professional.
PSO 2	Apply the knowledge of cyber-crime and cyber security.
PSO 3	Design and apply the suitable techniques, resources, modern methods, tools and technology to solve complex problems.
PSO 4	Inculcate professional and ethical principles.

Course Objectives (COB's)

COURSE OBJECTIVES	
COB1	Describe the core syntax and semantics of Python programming language
COB2	Illustrate the process of structuring the data using dictionaries, tuples and sets
COB3	Infer the Object-oriented Programming concepts in Python.

Course Outcomes (CO's)

At the end of the course, students will be able to

S.No	Course Outcome	Description	Bloom's Taxonomy Level
1	CO 1	Describe python language basics and flow control	Understand (L2)
2	CO 2	Use Data Structure like lists, dictionaries, tuples and sets to solve programming problems	Apply (L3)
3	CO 3	Articulate the Object-Oriented Programming concepts and Regular Expression.	Analyze (L4)
4	CO 4	Implement the basic methods of python modules like NumPy, Matplotlib	Analyze (L4)
5	CO 5	Apply the various libraries to analyze and visualize data.	Create (L6)

CO-PO/PSO Mapping:

Course Outcome	Bloom's Taxonomy Levels	Program Outcomes (PO's) Mapped												Program Specific Outcomes (PSO's)			
		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
														L3	L4	L6	L4
CO1	L2	2	1	3	-	2	-	-	-	2	-	-	-	3	1	3	2
CO2	L3	1	2	3	-	2	-	-	-	2	-	3	1	3	3	3	2
CO3	L4	-	1	2	1	-	-	-	-	2	-	3	1	3	3	3	2
CO4	L4	-	2	2	1	1	-	-	-	2	-	3	1	2	2	2	1
CO5	L6	-	1	2	-	-	-	-	-	3	-	3	1	2	2	2	1

Course Contents:

Module	Details	Tools Used / Assessment and Activity	CO Mapping	PO Mapping
I	INTRODUCTION TO PYTHON PROGRAMMING Overview-Applications -Print Statement-Writing and Executing a Python Program-Identifiers-Indentations- Keywords-Operators-Control Statements-Strings (Slicing- Indexing- Concatenation- other operations on Strings) - Functions	Activity: Online Certification Course – 13 to 15 Hours Coursera Certification	CO 1	PO1, PO4, PO5, PO6, PO7, PO9
II	DATA STRUCTURES Lists: list operations- list slices- list methods- list loop- mutability- aliasing- cloning lists- list Parameters-Slicing- Indexing- Concatenation- other operations on Sequence data type; Tuples: tuple assignment- tuple as return value; Dictionaries: operations and methods; advanced list processing – list comprehension; Sets: Creating a Set-Adding items to the set-Removing item from the set-Operations on Set	Assessment: Internal Test Activity: Coursera Certification , Flip Class/ Mini project	CO 2	PO1, PO2, PO3, PO4, PO5, PO7, PO9
III	OBJECT ORIENTED PROGRAMMING AND REGULAR EXPRESSIONS Classes: Classes and Instances-Inheritance— Polymorphism- Abstract Classes-Exceptional Handling- Regular Expressions using “re” module.	Activity: Coursera Certification & Mini project	CO 3	PO1, PO2, PO3, PO5, PO9, PO10
IV	INTRODUCTION TO NUMPY, PANDAS Computation on NumPy-Aggregations-Computation on Arrays-Comparisons, Masks and Boolean Arrays-Fancy Indexing-Sorting Arrays-Structured Data: NumPy’s Structured Array. Introduction to Pandas Objects-Data indexing and Selection-Operating on Data in	Activity: Coursera Certification & Presentation / Report writing, Mini project	CO 4	PO1, PO3, PO4, PO5, PO6, PO7, PO8, PO9,

	Pandas- Handling Missing Data-Hierarchical Indexing.			PO10
V	INTRODUCTION TO DATA ANALYSIS AND VISUALIZATION Overview of Data Analysis, Python Basics for Data Analysis and Visualization: Loading, Cleaning and Exploring the Dataset. Data Visualization using Matplotlib, Seaborn, Folium and Bokeh Library.(different types of charts and plots; line, area, histograms, bar, pie, box, scatter, and bubble, etc)	Assessment: Preparatory Exam Activity: Coursera Certification & Presentation / Report writing, Mini project	CO 5	PO1, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10,

Text Books (TB):

TB-1	Zhang.Y, An Introduction to Python and Computer Programming, Springer Publications, 2017
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Reference Books (RB):

RB-1	Wesely J.Chun, Core Python Application Programming, Prentice Hall, 3rd Edition, 2019.
RB-2	Mark Lutz , Programming Python, O'Reily Media Inc., 2019.
RB-3	T. R. Padmanabhan, Programming with Python, Springer Publications, 2019
RB-4	Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data", 2nd Edition, O'Reilly Media, 2020.
RB-5	Aurelien Geron, "Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems", 2nd Edition, O'Reilly Media, 2019. ISBN – 13: 978-9352139057.

Relevant Website (RW):

RW-1	https://www.w3schools.com/python/
RW-2	https://www.guru99.com/python-tutorials.html
RW-3	https://www.programiz.com/
RW-4	https://pythoninstitute.org

RW-5	https://docs.python.org/3/tutorial/index.html
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Video Links (VL):

VL-1	https://youtu.be/6yrsX752CWk
VL-2	https://youtu.be/ISItwlnF0eU
VL-3	https://youtu.be/YbkC8avAL60
VL-4	https://youtu.be/xlbSQJuHkRE
VL-5	https://youtu.be/0P7QnIQDBJY

MOOC Courses (MC):

Sl.No.	Platform	Topic	CO	Link	Duration
MC-1	Coursera	Introduction to Python Programming and Basic Concepts	CO1, CO2	https://www.coursera.org/videos/social-media-data-analytics/96A44	17 Mins
MC-2	Coursera	Expressions, Variables, and Python Operations	CO1	https://www.coursera.org/videos/python-for-applied-data-science-ai/gjVDL	2 hours, 23 minutes
MC-3	Coursera	Python Data Structures	CO2, CO3	https://www.coursera.org/programs/jgi-faculty-development-program-on-coursera-984mg/learn/python-data?source=search&collectionId=skill~python-programming-language	Approx. 18 hours
MC-4	Coursera	Python for Data Analysis: Pandas & NumPy	CO4	https://www.coursera.org/programs/jgi-faculty-development-program-on-coursera-984mg/projects/python-for-data-analysis-numpy?source=search	2 hours
MC-5	Coursera	Data Visualization with Python	CO4	https://www.coursera.org/programs/jgi-faculty-development-program-on-coursera-984mg/learn/python-for-data-visualization?source=search&collectionId=skill~python-programming-language	Approx. 20 hours

Total MOOC Course integration with Certification

60 Hours

Abbreviations & Expansions					
Pedagogy/Activity planned			Mode of Delivery		
P1	PPT presentation & Classroom interaction		M1	Synchronous – PPT	
P2	Blended Learning with Hands on		M2	Asynchronous	
P3	Flip Class & Module Quiz		M3	Synchronous - Hands on	
A1	Activity-1		M4	Synchronous - Discussion	
A2	Activity-2				

Session-wise Planning:

Module	Session	Topic	PPT	Readings and References	Pedagogy/ Activity planned	CO	Mode of delivery
01	1	INTRODUCTION TO PYTHON PROGRAMMING Overview-Applications	P MO...	TB1,TB2, RB1, OR1,VL1	P1	CO1	M1
	2	Print Statement			P1		M1
	3	Writing and Executing a Python Program			P1		M1
	4	Identifiers-Indentations			P1		M1
	5	Keywords-Operators			P1		M1
	6	Control Statements			P1		M1
	7	Strings (Slicing-Indexing)			P1		M1

	8-9	Concatenation- other operations on Strings) – Functions			P1		M1
02	10	Data structures : Lists: list operations List methods,List loop-list mutability-aliasing-cl oning lists		TB1,TB2, RB2, OR2,VL2	P1	CO2	M1
	11	List slices-list Parameters-Slicing Indexing-Concatenati on -other operations on Sequence Data type			P1		M1
	12	Advanced list processing – list comprehension;			P1		M1
	13	Tuples: Tuple assignment- Tuple Creation, tuple as return value, Basic Tuple operations			P1		M1
	14	Built-In Tuple Functions, List & Tuple Comparison.			P1		M1
	15	Dictionary: Operations, Loop through dictionary			P1		M1
	16	Dictionary Functions,Methods			P1		M1
	17	Sets: Creating a Set-Adding items to the set-Removing item			P1		M1

	18	Set Operations			P1		M1
03	19-20	OBJECT ORIENTED PROGRAMMING AND REGULAR EXPRESSIONS Classes: Classes and Instances-		TB2, RB2, OR3, VL3	P1	CO3	M1
	21	Inheritance			P1		M1
	22	Polymorphism			P1		M1
	23	Abstract Classes			P1		M1
	24-25	Exceptional Handling			P1		M1
	26-27	Regular Expressions using “re” module.			P1		M1
3504	28	INTRODUCTION TO NUMPY, PANDAS Computation on NumPy-Aggregations-		TB2, RB2, OR4, VL4, VL5	P1	CO4	M1
	29	Computation on Arrays- to .			P1		M1
	30	Comparisons, Masks and Boolean Arrays- -			P1		M1
	31	Fancy Indexing- Sorting Arrays-			P1		M1
	32	Structured Data: NumPy’s Structured Array.			P1		M1
	33	Introduction Pandas			P1		M1
	34	Objects-Data indexing and Selection-Operating on Data in Pandas			P1		M1

	35	Handling Missing Data			P1		M1
	36	Hierarchical Indexing			P1		M1
05	37	INTRODUCTION TO DATA ANALYSIS AND VISUALIZATION Overview of Data Analysis		TB3, RB3, OR5, VL5, V L6	P1	CO5	M1
	38	Python Basics for Data Analysis and Visualization:			P1		M1
	39	Loading, Cleaning and Exploring the Dataset.			P1		M1
	40	Data Visualization using Matplotlib			P1		M1
	41-42	Seaborn, Folium and Bokeh Library			P1		M1
	43-45	different types of charts and plots; line, area, histograms, bar, pie, box, scatter, and bubble, etc			P1		M1

Assessment Scheme: CA: UE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency	Weightage (%)	CO
1	Class Participation	Formative	Continuous	5	CO1, CO2, CO3, CO4, CO5
2	Activity-1	Formative	1	15	CO1, CO2, CO3, CO4, CO5
3	Activity-2		1	15	CO3
4.	Internal Test - 1		1	5	CO1, CO2, CO3
5	Internal Test - 2		1	10	CO3, CO4, CO5
6.	End Semester Exam	Summative	1	50	CO1, CO2, CO3, CO4, CO5
	Total			100	

Assessment Sheet with Rubrics for Grading & Evaluation

Rubrics: A1

Criteria	Poor Below Expectation	Fair Needs improvement	Good Meets Expectations	Excellent Exceeds Exceptions	Weightage
Static page design	Poor: Involved in project but No progress (1 Marks)	Moderate: no creativity involvement and completion of Partial webpage Design(2 Marks)	Good Execution: No creativity but on the completion of static web Page Design (3 Marks)	Very Good Execution: Creativity and Involvement of HTML/CSS/JS to design web page (4-5 Marks)	5
Dynamic page design	Poor: Involved in project but No progress (1 Marks)	Moderate: no creativity involvement and completion of Partial webpage Design(2 Marks)	Good Execution: No creativity but on the completion of Dynamic web Page Design (3 Marks)	Very Good Execution: Creativity and Involvement of PHP/MYSQL/ANGULAR to design web page (4-5 Marks)	5
Presentation	Answer less than 3 questions correctly (2 Marks)	Answer all 3 question correctly (3 Marks)	Answer 4 questions correctly (4 Marks)	Answer all 5 questions correctly (5 Marks)	5

Assessment Sheet with Rubrics for Grading & Evaluation

Activity-1 Report ON COURSERA CERTIFICATION

S. No.	USN No.	Student Name	Coursera Certificate	Coursera Content Learning	Viva	Total Marks
			5 Marks	5 Marks	5 Marks	15 Marks
1.						

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Activity-2 Mini Project

S. No.	USN No.	Student Name	Coursera Certificate	Database Connectivity process	Report	Total Marks
			5 Marks	5 Marks	5 Marks	15 Marks
1.						

Bloom's Taxonomy-Revised

LEVEL	DESCRIPTION	MEANING	ACTION VERBS
6	Creating	Can the student create a new product or POV?	Assemble, construct, create, change, combine, compose, design, develop, formulate, invent, modify, organize, propose, theorize, write
5	Evaluating	Can the student justify a stand or decision?	Appraise, agree, assess, argue, conclude, decide, defend, judge, prioritize, prove, rate, recommend, select, support, value
4	Analyzing	Can the student distinguish between different parts?	Contrast, compare, criticize, differentiate, discriminate, dissect, distinguish, examine, experiment, operate, question, simplify, test
3	Applying	Can the student use information in a new way?	Choose, demonstrate, dramatize, employ, illustrate, interpret, schedule, sketch, solve, use
2	Understanding	Can the student explain ideas and concepts?	Classify, describe, discuss, explain, identify, infer, locate, outline, paraphrase, recognize, report, summarize, select, translate
1	Remembering	Can the student recall or remember information?	Define, duplicate, find, list, label, match, memorize, name, omit, recall, repeat, state, spell, tell

Signature of the Faculty

Signature of the HoD